

SANDIP GUDLE

GRAPHICS PROGRAMMER

9637659641 | Pune - 411018 | sandipgudle121@gmail.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

SUMMARY

Computer Science graduate specializing in real-time rendering and low-level graphics systems. Built a custom C++ game engine from the ground up — shader pipeline, scene graph, and a real-time renderer in OpenGL/GLSL - alongside a 3D asset-import pipeline using Assimp. Comfortable working close to the GPU: shader authoring, texture, and material handling, and rendering performance optimization. Seeking an entry-level Graphics Programmer or Rendering Engineer role.

TECHNICAL SKILLS

Graphics APIs: OpenGL, OpenGL ES, DirectX11, GLSL

Languages: C++, C, C#, Python, JavaScript

Graphics & Engine Systems: Shader Pipelines, Scene/Model Rendering, Texture & Material Handling, Asset Import (Assimp), Dear ImGui Tooling

Core CS: Data Structures, Algorithms, Linear Algebra, OOP

Tools: Git, Visual Studio, VS Code, Win32 SDK

PROJECTS

Custom Game Engine - C++, OpenGL, GLSL, Dear ImGui

- Built a custom rendering engine from scratch, implementing shader management, texture, camera, and model rendering.
- Authored real-time rendering systems using **modern OpenGL and hand-written GLSL shaders**.
- Built in-engine editor and debugging tools using **Dear ImGui** to speed up iteration on rendering features.

3D Model Loading and Rendering System - C++, OpenGL, Assimp

- Built a **3D model import pipeline using Assimp** to load and render external model formats into the custom renderer.
- Implemented **mesh processing, texture loading, and material handling** for real-time rendering in OpenGL.
- Optimized rendering of complex 3D assets through **custom shader programs**.

OpenGL Graphics Application - C++, OpenGL, Win32 SDK

- Developed a graphics application rendering 2D and 3D geometric primitives with **texture mapping and lighting effects**.
- Built a custom UI and menu system directly on the **Win32 API**, without a game engine or framework.

EXPERIENCE

Graphics Programming Trainee - AstroMediComp 2023-2025

- Collaborated in a team to develop graphics programming projects using **Modern OpenGL, GLSL, and C++**.
- Worked on implementing rendering features including **shader programming, model rendering, texture mapping, and lighting**.
- Used **Git** for version control, feature development, **code integration**, and collaborative workflows.
- Participated in code reviews, debugging, and testing to ensure stable rendering functionality.
- Developed and tested graphics applications across **Windows, Linux, macOS, Android, iOS, and WebGL**, gaining hands-on experience with cross-platform rendering workflows.

EDUCATION

M.Sc. Information Technology - Dr. DY Patil ACS College - 9 CGPA (2024–2026)

B.Sc. Computer Science - SRTM University - 9 CGPA (2021–2024)

CERTIFICATIONS & SEMINARS

- **Real-Time Rendering in OpenGL and DirectX** - AstromediComp
- **Heterogeneous Parallel Programming (CUDA & OpenCL)** – AstroMediComp
- **Multi OS Installation Seminar** - AstromediComp